

Connections AI Solver

Studienarbeit

for the Informatik Department at the
DHBW Heidenheim

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TODO. TODO 2026

Processing period	January – March 2026, July – September 2026
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Course	Allgemeine Informatik
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Declaration of Independence

I hereby declare that I have written my study on the topic “*Connections AI Solver*” independently and that I have not used any sources or aids other than the indicated ones. I also declare that the submitted electronic version matches the printed version.

MyPlace, August 9, 2026

Place, Date

Signature

Abstract

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1 Introduction

NYT Connections asks players to sort sixteen words into four groups of four. Each group carries one of four difficulty tiers, ranging from yellow (easiest) to purple (hardest). Published daily, the puzzle has become a popular benchmark for testing the reasoning abilities of language models. Two challenges make Connections difficult to solve.

The first challenge is lexical or semantical. Solving Connections requires several distinct kinds of language understanding at once. Some groups rest on straightforward synonymy. Others depend on shared themes, wordplay, or knowledge of pop culture and named entities. A single puzzle typically mixes several of these relation types, and is deliberately constructed so that some words could plausibly belong to more than one candidate group under a superficial reading, an effect commonly referred to as a red herring.

The second challenge is structural. Because a word cannot be reused across groups, correctly identifying even one group depends not only on recognizing a plausible shared category among four words in isolation, but also on the arrangement of the remaining words. A locally plausible grouping can still be wrong if it leaves the remaining words unable to form additional valid groups of four.

In previous studies, the first challenge has been widely discussed, through embedding-based clustering or large language models to propose or judge candidate groupings Section 3.1. This thesis evaluates how far such approaches can be pushed, including several forms of knowledge injection beyond raw embeddings, and finds that they consistently fall short in similar ways (note: to be written, do the reference later!!!). The second challenge, ensuring that all sixteen words are partitioned simultaneously and exclusively, has received comparatively little attention on its own. Taking inspiration from solving Wordle algorithmically with information theory [1], this thesis treats the puzzle’s interactive structure, submitting a guess, observing feedback, and revising, as central to addressing this second challenge, and asks how much of the puzzle’s difficulty it can resolve by algorithm.

1.1 Structure

2 Related Work

3 Related Work

This chapter provides an overview of prior work relevant to this thesis, including work that directly solves Connections using LLMs and work that inspired the methods used in this study. It first introduces prior work on solving Connections with AI. It then presents related work on the methods used, which serves as the knowledge base for this study.

3.1 Prior Work on Connections Solving

Current work on this topic mainly uses LLMs as the main solution for solving Connections. These studies compare how well different models solve Connections. Some use Connections as a benchmark to test LLM capability, while others explore different prompting techniques and methods. Todd et al. use three methods, namely a sentence-embedding baseline, a comparison between two LLMs (GPT-3.5-Turbo-1106 and GPT-4-1106-Preview), and Chain-of-Thought (CoT) prompting. Their experiments allow five incorrect and five invalid guesses per puzzle, which departs from the original game’s rules. GPT-4 Turbo with CoT achieves the highest success rate at 58.11%. The best-performing sentence-embedding baseline (MPNet) reaches 11.6%. LLMs are often stumped by categories that involve non-semantic word properties, abstract features, or context-dependent usage. They also report a finding relevant to this thesis. Puzzle order matters, since presenting puzzles in their original, grouped order achieves substantially higher accuracy than presenting them in randomized order [2].

A similar work from Samadarshi et al. benchmarked several LLMs against both novice and expert human players and creates a taxonomy of knowledge types required to solve connections. They used in total 438 Connections puzzles and their best performing models Claude Sonnet 3.5 achieved 18% success rate of fully solving the puzzles. They categorizes NYT Connections categories along two main dimensions, word form (covering morphology, orthography, phonology, and multiword expressions) and word meaning (covering semantic relations such as synonymy, polysemy, and hypernymy, along with associative and encyclopedic relations), with a further category for groups that combine both form-based and meaning-based cues. Results show that models performed comparably well on semantic relation categories but significantly worse on the other types.

3 Related Work

More recent work from Loredó Lopez, McDonald, and Emami compare three game configurations: single attempt, four tries without feedback (one away prompt) and four tries with feedback. Human solvers outperform LLM under all three configurations. The best performing model Claude 3.5 under full hints reaches 40.25%, while the human full-hints score reaches 60.67%. They also experiment on different prompting techniques. CoT and self-consistency prompting yield diminishing or even negative returns as puzzle difficulty increases. Simple input-output prompting sometimes outperforms these more elaborate reasoning strategies. Their heuristic (non-LLM) baseline, using Multilingual-E5-Large-Instruct embedding and beam search, reaches 13.25% accuracy.

Zhang, Jiang, and Ye compare in-context prompting of a large model (DBRX) against fine-tuning and distilling a smaller model (Mistral 7B) on solved puzzles. They found that fine-tuning a smaller model (Mistral 7B) directly on solved puzzles fails to capture the task itself, which underperformed even their clustering baseline. Their strongest result comes from Verbal Reinforcement Learning, in which a larger supervisor model critiques the fine-tuned model’s output in natural language across three rounds, roughly doubling the average per-puzzle match rate relative to single-pass prompting [5].

3.2 Word Puzzles as Weighted Constraint Satisfaction Problem (CSP)

CSPs are mathematical questions defined as a set of objects whose state must satisfy a number of constraints or limitations [6]. A (finite domain) CSP can be expressed in the following form: given a set of variables, together with a finite set of possible values that can be assigned to each variable, and a list of constraints, find values of the variables that satisfy every constraint [7]. It is formally defined as a triple $\langle X, D, C \rangle$ where X is a set of variables, D specifies a domain of possible values for each variable, and C is a set of constraints restricting which combinations of value assignments are permitted [6].

In artificial intelligence, a Weighted Constraint Satisfaction Problem, also known as Valued Constraint Satisfaction Problem, is a constraint satisfaction problem in which preferences among solutions can be expressed [8]. A plain CSP, treats every solution as equally good: it either satisfies the constraints or it does not. Connections needs more than that. Not every four-groups-of-four split is a correct solution, only one is. What is available

instead is a measure of how well four words seem to belong together, based on their similarity. Thus, Connections can be defined as a weighted CSP. It keeps the hard constraint (a valid four-groups-of-four partition) but adds a score to each candidate value assignment, so that solving the problem means finding the constraint-satisfying assignment with the best total score, not just any assignment that satisfies the constraints.

3.2.1 Solving Word Puzzles as Weighted CSP

Applying this concept to word puzzles is not new. Ginsberg formalizes American-style crossword solving as a weighted CSP in the program DrFill: each grid cell is a variable, each candidate fill is a value, and the constraints require intersecting words to agree on shared letters [9]. Because many fills are plausible for a given clue, each candidate fill is given a likelihood-based weight, and Dr.Fill searches for the assignment that maximizes total weight while still satisfying the grid's hard constraints. It draws candidate fills from several sources: a database of over 47,000 previously published crosswords with 1.9 million unique clues, two dictionaries of varying size, 1.2 million synonyms drawn from WordNet and other online sources, and a database of Wikipedia titles and word pairs. Each candidate fill is then scored on five criteria: whether the clue matches one seen before in the clue database, whether the fill's part of speech agrees with the part of speech inferred from the clue, the fill's crossword "merit" (a hand-rated measure of how desirable a word is as a crossword answer in general), whether the fill is an abbreviation, and whether the clue is a fill-in-the-blank type resolvable against Wikipedia text. Dr.Fill then searches for the assignment that maximizes total score while still satisfying the grid's hard letter-intersection constraints. Two further design choices in Dr.Fill are worth noting. First, they found that looking further ahead before committing to a choice gave no real benefit, and the system used only a single step of lookahead instead. Second, branch-and-bound, a standard pruning technique for weighted CSPs, was tested and dropped because it added little value in this domain, as a result postprocessing was used instead. In both cases, a more sophisticated search method did not beat a simpler one, once the scoring function and hard constraints were doing most of the work. Since complete knowledge of what each clue means is impractical to obtain, Dr.Fill relies mainly on the hard constraint that crossing words must share letters to narrow down candidates, not on understanding the clues themselves. Dr.Fill inspired the methodology used in this thesis, which also

treats Connections as a weighted CSP and uses a scoring function to evaluate candidate groupings.

3.2.2 Set Partitioning

The constraint C of Connections (every word belongs to exactly one group and every group contains exactly four words), more precisely speaking, is a combinatorial optimization problem known as set partitioning. A partition of a set is a way of dividing all of its elements into non-empty, pairwise disjoint subsets whose union is the original set [10]. The set partitioning problem asks, given a ground set and a collection of candidate subsets each associated with a cost, to select a subset of these candidates that forms a partition of the ground set at minimum or maximum total cost [11]. The sixteen Connections puzzle words are the ground set, candidate four-word groups are subsets among which a selection must be made, and each candidate group’s score from the weighted CSP explained above serving is its cost. Identifying this structure is what motivates the grouping strategy adopted later in this thesis.

Formulating a puzzle as a weighted CSP defines what to search for. Among all groupings that satisfy the hard constraint, the goal is the one with the highest total score. This does not guarantee that the grouping is correct. The hard constraint only rules out invalid partitions. Many valid partitions remain, and the scoring function is what distinguishes them from each other. If the scoring function is imperfect, the highest-scoring grouping can still be wrong. Puzzle games like Connections give players a further source of information, namely the response the game returns after each attempt. Using this response turns the problem from a single optimization into a sequence of attempts, where each attempt narrows down what remains possible. Section 3.3 covers how much a single response is worth, and Section 3.4 covers what to do with it.

3.3 Information Theory

Information theory is the mathematical study of the quantification, storage, and communication of a particular type of mathematically defined information [12]. It gives information a precise mathematical definition grounded in probability [13]. Gaining information and reducing uncertainty about a random event are the same thing. By convention, information is measured in base-2 logarithms, giving the bit as its unit: each time half of the

3 Related Work

probability mass is ruled out, exactly one bit is gained [14]. The information contained in an outcome with probability $p(x)$ is $\log_2 \frac{1}{p(x)}$.

The probability-weighted average of this information across all outcomes is the entropy, denoted $H(X)$:

$$H(X) = \sum_{x \in \mathcal{X}} p(x) \log_2 \frac{1}{p(x)}. \quad (3.1)$$

Entropy serves as the connection between probability and information content. It measures how uncertain a random event is on average. Higher entropy means more uncertainty to begin with, so more information must be acquired to become certain about the outcome. Entropy is maximized when probability is spread evenly over all outcomes. In that case it reduces to

$$H_{\max}(\mathcal{X}) = \log_2 |\mathcal{X}|, \quad (3.2)$$

where $|\mathcal{X}|$ is the number of possible outcomes [14].

Conditional entropy, $H(X | y)$, measures how much uncertainty about X remains after observing an outcome y . The reduction in entropy achieved by an observation is the information gain. Taken in expectation over all possible outcomes of an observation, before that observation is actually made, this gives the expected information gain of an action:

$$IG(\text{guess}) = H(X) - \mathbb{E}[H(X | \text{feedback})]. \quad (3.3)$$

This quantity is what solvers use to decide which action is most useful before knowing what the feedback will be. An action with higher expected information gain is expected to narrow down the remaining candidates more, regardless of which specific feedback is eventually returned.

3.3.1 Solving Wordle with Information Theory

Wordle is another word puzzle game from NY Times. It asks the player to guess a five-letter word within six attempts. After each guess, every letter is marked as correct and in the right position, present but in the wrong position, or absent, giving detailed per-letter feedback. Information theory based Wordle solvers apply this framework directly: at each step, the candidate word maximizing expected information gain is chosen as the next guess [15]. According to Grant Sanderson pointed out in his video, the best possible first guess yields an expected information gain of 5.89 bits, reducing the remaining uncertainty by

roughly half in a single observation [1]. Aladaileh et al. report that entropy-based selection outperforms a heuristic that selects words by letter distribution, and propose the approach as a general baseline for applying Shannon entropy to other games [15]. Information theory can be applied to Connections as well. Detailed description see next chapter.

3.4 Strategies for Puzzle Games

Section 3.2 introduces the structural constraint, and then section 3.3 as a way of measuring how much a single observation can reveal. After measuring information, a solver needs a rule that turns the current state of knowledge into the next move. This section covers the other question: given feedback from previous attempts, how should the next guess be chosen?

Cunanan and Thielscher provide a solution for Wordle and for a wider class of games with the same structure. After a sequence of guesses and responses, the candidate set is the set of possible secrets that remain consistent with every response observed so far. A strategy is a rule that maps the current candidate set to the next guess, and different strategies can be compared by scoring every possible guess against the candidate set.

Several such scoring functions were developed earlier for Mastermind. These functions include minimizing the size of the largest candidate set that could remain, maximizing information gain as described in the previous section, and maximizing the number of distinct responses a guess can produce [17]. Cunanan and Thielscher test these functions on Wordle and find that none performs best on its own. Maximizing information gain in particular does not yield the optimal strategy [16].

4 Problem Formation

The previous Chapter Chapter 3 introduced three foundational concepts that apply to Connections. Firstly, it can be treated as weighted CSP and its grouping requirement can be concluded as a set partition problem. Secondly, information theory provide a framework to measure how much remains unknown about a puzzle and how much a single observation can reveal. Thirdly, after quantifying the information and uncertainty, connections provides a feedback mechanism that can be used to determine the next answer.

This chapter starts with Section 4.1, which states the Connections game rules and defines the two evaluation settings used throughout this thesis. Section 4.2 expresses the puzzle as a weighted CSP mathematically. Section 4.3 counts how many candidate solutions exist and how much uncertainty that represents. Section 4.4 then quantifies how much of that uncertainty the game’s feedback can remove.

4.1 Problem Setting

A Connections puzzle presents sixteen words that must be split into four groups of four. Each group corresponds to a hidden category, and every word belongs to exactly one group. The player does not submit a full solution at once. Instead, one group of four words is submitted at a time, and the game responds before the next submission is made.

Three responses are possible. If all four submitted words indeed belong to one category, 4 words will be placed on the same row and the category will be revealed. If exactly three of them do, player will be prompted with one away. Otherwise the response is wrong. Both one-away and wrong count as mistakes, and the game ends in failure after four mistakes. Correct submissions do not consume the mistake budget. A correct submission takes those four words out of consideration for the remaining groups. The puzzle therefore shrinks as it is solved, and each correct answer both reduces the number of remaining possibilities and simplifies the problem that remains.

There are four levels of difficulty, starting with yellow group (the easiest), then green, blue, and purple (the hardest), which reflect the intended difficulty in recognizing the underlying grouping relation. So if player are able to solve the easy groups first, the remaining puzzle will be easier to solve.

Table 4.1: Relation category for NYT Connections with illustrative examples.

Relation type	Example category + words
Semantic similarity	<i>FURTHEST POINT</i> : END, EXTREME, OPPOSITE, POLE [19]
Semantic relatedness	<i>THINGS YOU CAN DO ON SOCIAL MEDIA</i> : COMMENT, LIKE, LURK, POST [19]
Form- or rule-based relations	<i>ART MOVEMENTS, WITH “-ISM”</i> : BRUTAL, IMPRESSION, MANNER, REAL [19]
Referential relations	<i>RIHANNA #1 HITS</i> : DIAMONDS, SOS, UMBRELLA, WORK [20]
Mixed categories	<i>CAR BRAND HOMOPHONES</i> : INFINITY, MINNIE, OPAL, OUTIE [21]

The group relations are diverse. Some categories are driven by semantic similarity, such as near-synonymy or membership under a shared hypernym (e.g., “types of flowers”). Others reflect semantic relatedness: words are not strictly similar, but cohere within a common theme or script (e.g., actions in a typical scenario). A third class consists of form- / rule-based relations, where membership is determined by surface regularities such as spelling or word-formation patterns (orthography, morphology), or by conventional meanings of symbols and abbreviations. Referential relations require external world knowledge about named entities or factual lists (e.g., trivia- or culture-dependent categories).

Notably, the hardest (purple) categories often draw on less standard relations such as word and letter patterns (e.g., rhymes, homophones, anagrams, palindromes, or unmarked fill-in-the-blank phrases) and may also involve pop-culture or proper-noun knowledge. [18] As a result, they frequently exhibit mixed relations that combine more than one relation type and cannot be explained well by a single type alone.

An overview of these relation types and representative examples is provided in Table 4.1.

This categorisation is derived from inspecting published puzzles, and it broadly aligns with the taxonomy proposed by Samadarshi et al., which concentrates on the kinds of

knowledge a LLM must possess [3], while ours classifies the relation that holds a group together. The single semantic category they proposed was split into similarity and relatedness in this taxonomy, since the two behave differently under embedding-based scoring, and multiword expressions was also no longer treated as an individual category.

4.1.1 Evaluation Settings

This thesis evaluates solvers under two settings, and the distinction between them matters throughout. In the **one-shot** setting, a solver commits to a complete partition of all sixteen words in a single pass, receiving no feedback at any point. The same setting is also used by Todd et al. [2], Samadarshi et al. [3], and Loredó Lopez, McDonald, and Emami [4]. In the **interactive** setting, the solver submits one group at a time, observes the response, and may revise before submitting again, subject to the four-mistake budget, which is also adopted by Loredó Lopez, McDonald, and Emami [4]. The second reflects how the puzzle is actually played.

The one-shot setting isolates what is being measured. With no feedback available, the outcome depends entirely on the scoring function, so any change in performance can be attributed to whatever was changed in that function. Under the interactive setting the two are entangled: a weak scoring function may be rescued by feedback, and a strong one may be squandered by a poor choice of what to submit first, leaving it unclear which is responsible for the result. The interactive setting measures whether a puzzle is won under the rules. The two settings pose different problems and are measured accordingly. The metrics used for each are given in Chapter 5.

4.2 Connections as a Weighted CSP

The previous section described the puzzle in words. This section restates it in the form introduced in Section 3.2, where a constraint satisfaction problem is given as a triple $\langle X, D, C \rangle$ of variables, domains and constraints, and a weighted variant adds a score to each candidate assignment so that solutions can be ranked rather than merely accepted or rejected. Both parts are needed here. The constraint captures what makes a partition legal, and the score captures what makes one legal partition more plausible than another.



Figure 4.1: The NYT Connections puzzle answer of 12 June 2023.

Connections maps onto the triple $\langle X, D, C \rangle$ directly. Let $W = \{w_1, \dots, w_{16}\}$ be the sixteen puzzle words. Each word gets its own variable, so $X = \{x_1, \dots, x_{16}\}$. The variable x_i does not stand for the word itself, but for the decision of which group w_i is placed in. Every variable has the same domain, $D_i = \{1, 2, 3, 4\}$, one value for each of the four groups.

Take the puzzle of 12 June 2023 as an example (Figure 4.1 shows the answer). Its sixteen words are BUCKS, HAIL, HEAT, JAZZ, KAYAK, LEVEL, MOM, NETS, OPTION, RACECAR, RAIN, RETURN, SHIFT, SLEET, SNOW and TAB. Numbering its sixteen words gives $w_1 = \text{BUCKS}$, $w_2 = \text{HAIL}$, $w_3 = \text{HEAT}$, and so on up to $w_{16} = \text{TAB}$. Placing BUCKS in group 2 means setting $x_1 = 2$, and placing JAZZ there as well means $x_4 = 2$. A complete assignment $x = (x_1, \dots, x_{16})$ records one such decision per word and therefore describes one way of dividing the board. One important aspect is that the group numbers themselves carry no special meaning for the solver in this thesis. They serve only to record which words end up together, and any consistent relabelling describes the same division of the board. A solver is not asked which group corresponds to which, nor to identify the difficulty level a group belongs to, only which four words go together.

4 Problem Formation

Let k be a group number, so $k \in \{1, 2, 3, 4\}$, and write G_k for the set of words that have been assigned to it:

$$G_k = \{w_i : x_i = k\}. \quad (4.1)$$

The condition $x_i = k$ selects those indices i whose word was placed in group k , and G_k collects the corresponding words. In the assignment described above, $x_1 = 2$ and $x_3 = 2$, so BUCKS and HEAT both belong to G_2 . Every assignment x determines all four sets $G_1, \dots, G_4$ in this way. The two notations describe the same thing from different directions: x records, for each word, which group it went to, while G_k records, for each group, which words it received.

The constraint set C then requires each group to hold exactly four words:

$$C : \quad |G_k| = 4 \quad \text{for every } k \in \{1, 2, 3, 4\}. \quad (4.2)$$

Satisfying Equation 4.2 is not difficult, but only one partition recovers the puzzle's intended grouping. The constraint treats all valid partitions as equally acceptable and we need something to distinguish the correct one, and that role is filled by a scoring function.

Let $s(g)$ be a function that takes a group of four words $g \subset W$ and returns a score or weight for how plausibly they belong to a common category, with higher values indicating stronger evidence. How s is constructed in detail is explained in Chapter 5. The problem is then to find

$$x^* = \arg \max_x \sum_{k=1}^4 s(G_k) \quad \text{subject to } |G_k| = 4 \text{ for all } k. \quad (4.3)$$

The assignment x^* is the one scoring highest among all assignments satisfying the constraint. It is worth being explicit that x^* is not by definition the correct answer, but the partition that s ranks highest, and whether that coincides with the puzzle's intended grouping depends entirely on how well s reflects the relations the puzzle is built on. Where s is imperfect, the correct partition may score below an incorrect one, and no amount of searching will recover it, because the information needed to prefer it was never encoded in the score. Much of this thesis is concerned with how close s can be brought to that standard, and with what happens when it falls short.

The sum in Equation 4.3 runs over all four groups, so an assignment is judged by the combined plausibility of the whole partition and not of any group on its own. The 12 June 2023 puzzle shows why this matters. HEAT sits naturally alongside HAIL, RAIN, SLEET and SNOW, and a solver scoring that group in isolation would find it convincing. The group is nonetheless wrong: HEAT belongs to NBA TEAMS, and committing to a weather group containing it leaves BUCKS, JAZZ and NETS without a fourth member. A locally plausible group can only be accepted if the twelve words it leaves behind can still form three groups that score well, and the summation over all four groups is what enforces that condition.

4.3 The Hypothesis Space

4.4 Information from Feedback

5 Methodology

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